

System Design

Library Management System

1. ER Diagram (Entity Relationship Diagram)

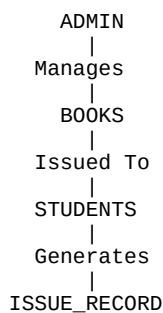
Entities

- Admin
 - Admin_ID
 - Name
 - Password
- Books
 - Book_ID
 - Book_Name
 - Author_Name
 - Quantity
- Students
 - Student_ID
 - Student_Name
 - Course
- Issue_Record
 - Issue_ID
 - Student_ID
 - Book_ID
 - Issue_Date
 - Return_Date

Relationships

- Admin manages Books
- Students issue Books
- Issue_Record stores issue and return details

ER Diagram Structure



2. DFD (Data Flow Diagram)

Level 0 DFD

Input

- Student details
- Book details

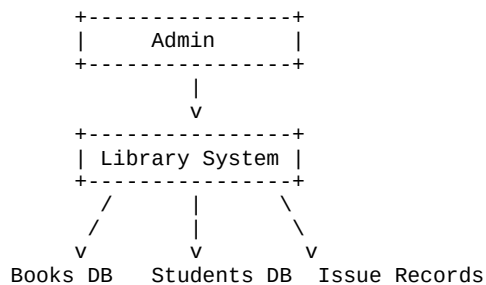
Process

- Add books
- Search books
- Issue books
- Return books

Output

- Book availability
- Issue records
- Return records

DFD Structure



3. Use Case Diagram

Actors

1. Admin
2. Student

Use Cases

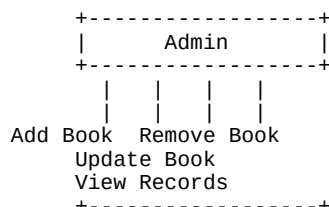
Admin

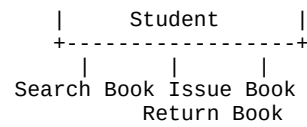
- Add Book
- Remove Book
- Update Book
- View Records

Student

- Search Book
- Issue Book
- Return Book

Use Case Diagram Structure





Conclusion

The system design of the Library Management System explains how data flows inside the system and how users interact with different features. ER Diagram, DFD, and Use Case Diagram help in understanding the structure and working of the project clearly.